1 Chapter 2.3

Problem 13. *Give a combinatorial proof: If $n \geq 1$ and $k \geq 1$, then*

$$S(n,k) = \sum_{j=1}^{n} \binom{n-1}{j-1} S(n-j, k-1)$$

Answer:
Testing the water of Lake Minnetonka.

2 Chapter 2.4

Problem 1. *You have 40 pieces of candy to distribute among 10 children. Find the number of ways to do this in each of the following situations. Leave your answers in standard notation.*

(a) *The pieces of candy are different and each child gets at least one piece.*

(b) *The pieces of candy are the same and each child get any number of pieces.*

(c) *The pieces of candy are different but you distribute them among 10 of the same paper bags.*

(d) *The candy is the same you distribute them in 10 bags that are the same and each bag contains at least one piece.*

(e) *The candy is different and each child gets exactly one pieces, so there is some left over candy.*

(f) *The candy is different and Frank receives four pieces.*

Answer:

Problem 4. *Use type vectors to establish the bijection in Theorem 2.4.2.*

Theorem 1. *If $n \geq 1$ and $k \geq 1$, then:*

$$P(n,k) = \sum_{j=1}^{k} P(n-k, j)$$

Answer:

Problem 7. *Under what conditions on n and k is the statement $P(n,k) = P(n-1, k-1)$ true?*

Answer:

Problem 8. *Give a bijective proof of the following: The number of partitions of n is equal to the number of partitions of $2n$ into n parts.*

Answer: